

Emir Gicić

📍 Belgrade, Serbia ✉ emirgicicnp@gmail.com ☎ +381621509906 🔗 in/emir-gicic-47672a309 🌐 github.com/GileEmir

SUMMARY

Diligent and analytical software developer with hands-on experience in creating full-stack applications, compilers, and embedded systems. Proficient in multiple programming languages, including C/C++, Python, Java, and TypeScript with professional experience in web development with Angular. Skilled in crafting effective solutions across both front-end and back-end development, with a strong commitment to learning and integrating new technologies.

PROJECTS

Assembler, Linker, and Emulator

August 2024 - September 2024

- Achieved full compatibility of the toolchain for Linux environments.
- Created a toolchain with an assembler, linker, and emulator for a 32-bit, two-address processor..
- Used Flex and Bison to simplify code translation with lexical analysis..
- Technologies used: C\C++, Flex & Bison.

Full-Stack Web Application for Exterior Decoration - (available on github.com/GileEmir)

August 2024 - September 2024

- Developed a web platform, "Your Imagination – Your Garden," to help users design and manage garden decorations for private residences and restaurants.
- Implemented role-based functionalities, allowing garden owners to manage projects and collaborate with decorators, while decorators could apply for tasks and update garden designs.
- Utilized Angular 16 for the frontend and Express/Node.js for the backend.
- Integrated MongoDB as the non-relational database for managing user and project data.
- Technologies used: Angular, Node, HTML, CSS.

RISC-V multithreaded kernel - (available on github.com/GileEmir)

June 2024 - July 2024

- Developed a RISC-V supervisor-level kernel with memory allocation and threading support.
- Added multithreading capabilities, enabling the kernel to manage multiple tasks concurrently.
- Designed and integrated round-robin scheduling for task management.
- Technologies used: C\C++, GNU RISC-V assembly.

MicroJavaCompiler - (available on github.com/GileEmir)

October 2024 -February 2025

- Developed a compiler for MicroJava programming language, including lexical, syntax and semantic analysis, and code generation.
- Utilized Flex for lexical analysis and Employed CUP for syntax analysis to parse tokens and construct an abstract syntax tree (AST).
- Implemented semantic analysis to ensure type safety and correct variable usage.
- Generated bytecode from the AST and used mj-runtime-1.1.jar for running the bytecode.
- Technologies used: Java, Flex & CUP.

EXPERIENCE

Tourism Software Solutions – TSS in/company/tourism-software-solutions-tss

Software Developer at TSS • Mentored by Haris Zenovic [\[in/haris-zenovic-47742a131\]](https://in/haris-zenovic-47742a131)

- worked on various frontend framework as well as backend, additionally implemented docker images for different environments

EDUCATION

Bachelor of Science - Electrical Engineering

University of Belgrade • Faculty of Electrical Engineering module Computer Science

High School

Novi Pazar Gymnasium • Novi Pazar • Sep 2017 - Jun 2021

SKILLS

C/C++, Java, Python, SQL, TypeScript, JavaScript, HTML, CSS, MongoDB, Express, Angular, Node, Git, Flex, Bison, SpringBoot, Verilog

Languages: Serbian (native), English (advanced)